
Emergent vs. Supervised Representations of Outcome-Proximity in a King+Pawn-vs-King Move-Prediction Transformer

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Abstract

The objective of this project is to study if a small Transformer, trained only to predict the next move in King+Pawn-vs-King (KPK) chess endgames, develops an internal notion of how close a position is to its outcome, using a Syzygy tablebase as our ground truth, rather than the noisy human games used by Chess-GPT (Karvonen, 2024). We find a partial answer going through five experiments, which include static probing, outcome probing, out-of-distribution generalization, activation patching and its tactical-criticality stratification, plus two ablations: when the model is trained only for the next-move, it develops a weak and layer-distributed representation of outcome-proximity, with no single layer directly responsible for it. This representation becomes concentrated into a single layer (median causal recovery 1.00 vs. 0.07–0.21) when we add the auxiliary training objective of predicting win/draw/loss and distance-to-outcome, but it makes it generalize worse outside the distance range seen in training. A hand-built geometric-feature baseline beats the unsupervised model on static outcome classification, but loses clearly on distance generalization. Finally a KRK (King+Rook) fine-tuned merges with it through a monotonic accuracy tradeoff, rather than a disastrous collapse; we deduce that the shared vocabulary and initialization design is compatible for merging.

1. Introduction

World models are fundamental for recreating environments into representations useful for planning (Ha & Schmidhuber, 2018). In particular in the context of game-playing Transformers, several works found that models

trained only to predict the next move develop an internal board-state representation decodable with linear probes: it happens with Othello-GPT (Li et al., 2023) and Chess-GPT (Karvonen, 2024). With this project we ask an harder question: does such model also represent how close the game is to being decided? A quantity that requires implicit look-ahead, not only tracking the current position. I used linear probes (Alain & Bengio, 2016) as the main tool, but I also took into consideration Belinkov (2022)’s warning that probe accuracy confuses decodability with causal use. For this reason M5 tests causal use directly via activation patching. I focused particularly on KPK rather than KRK: KRK is solved usually by a mechanical and trivial procedure, while KPK has more strategic depth, thanks to opposition, zugzwang and key squares, where a single tempo flips the outcome. The paper’s central methodological choice was the 3-piece Syzygy tablebase, which gives the exact WDL and distance-to-zeroing (DTZ) value for every position at no cost, unlike Chess-GPT’s noisy human-game. The move vocabulary (Section 2) was deliberately built shared across KPK and KRK from the start of the project, in order to anticipate a weight-merging experiment (M6, Section 4).

2. Related Work

Linear probing was formalized by Alain & Bengio (2016) and surveyed critically by Belinkov (2022). The causal tracing methodology behind M5 follows Meng et al. (2022); Wang et al. (2023); Elhage et al. (2021). The multi-task ablation follows Caruana (1997); the weight-merging experiment (M6) has the task-arithmetic framing of Ilharco et al. (2023). The base architecture is the standard decoder-only Transformer (Vaswani et al., 2017).

3. Method

Data Generation. We self-play KPK games (python-chess (Fiekas, 2024) with Syzygy (de Man, 2023)) with different policies: with probability 0.85 the move maximizes/minimizes tablebase DTZ (near-optimal), otherwise we have a uniform-random legal move. 75% of starting positions force the defending king within 1

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square of the pawn, which we verified raises the true zugzwang rate (flipping the side to move, flips the winner) from 19% to 57%, targeting tactically rich positions. We generated 200k games (mean length 54 plies) and we had six automated correctness checks (tokenizability, DTZ/WDL consistency, absence of non-progressing move cycles).

Model. A 6-layer, 4-head causal Transformer ($d = 256$, 5.7M params) predicts the next move from UCI-style move tokens. I trained 25 epochs, where it reaches 81.2% top-1 accuracy, close to the ~ 85 –87% ceiling implied by the 15% random-move training noise.

Probing (M2/M3). Linear probes on frozen hidden states at every layer decode king/pawn/rook squares (M2) and WDL / a 6-bucket DTZ magnitude (M3), the latter class-weighted against a 66/24/10 draw/win/loss imbalance.

Generalization (M4). A regression probe predicts raw $|\text{DTZ}|$, trained *only* on near-outcome positions ($|\text{DTZ}| \leq 5$) and evaluated on far positions never seen in that magnitude range ($|\text{DTZ}| \geq 10$), against a constant-prediction baseline. We repeat with a 2-layer MLP as a nonlinear control, to separate not linearly readable from not present at all.

Causal patching (M5). In this phase I isolated one layer marginal contribution to the residual downstream ($\Delta h_L = h_L - h_{L-1}$) and injected it from the clean to the corrupt run, to measure recovering of a fixed downstream WDL-probe readout for 150 minimal pairs (identical move prefix, differing only in a final move that preserves or worsens the WDL outcome, near conversion). Otherwise the naive full residual stream patching probably would have recovered 100% for any single-token-difference pair, regardless of where the information lives.

Ablations and M6. *Multitask:* two auxiliary heads (WDL, DTZ-bucket) on the last layer, trained jointly with the next-move loss (Caruana, 1997) from a fresh initialization, isolating the training-objective effect alone. *Geometric baseline:* identical M3/M4 probes on 11 hand-computed features (king/pawn squares, pairwise distances, distance-to-promotion, side to move) instead of hidden states. *M6:* a KRK specialist is fine-tuned starting from the base KPK checkpoint’s weights (not independent initialization), making a task-vector merge (Ilharco et al., 2023) well-defined: with $\tau = \theta_{\text{ft}} - \theta_{\text{base}}$, we evaluate $\theta(\alpha) = \theta_{\text{base}} + \alpha\tau$ on held-out KPK and KRK accuracy across $\alpha \in \{0, 0.25, 0.5, 0.75, 1.0\}$.

Table 1. M2/M3 probing accuracy by layer, vanilla model (mean of 2 seeds). King squares saturate almost immediately; WDL/DTZ peak at layers 3–4 and degrade by layer 5.

Layer	own-K	other-K	pawn	WDL	DTZ
0	0.88	0.93	0.19	0.53	0.39
2	0.93	0.94	0.57	0.71	0.59
3	0.93	0.94	0.60	0.74	0.63
4	0.93	0.94	0.62	0.74	0.62
5	0.93	0.94	0.63	0.71	0.58

Table 2. Comparison across the three model variants with best layer for each metric.

	WDL F1	M4 (MLP)	M5 (L5)
Vanilla	0.74	+1.70	0.07–0.21
Multitask	0.93	+1.39	1.00
Geometric baseline	0.82	+0.08	

4. Results

M2 (Table 1): both kings are decodable almost immediately (88%/93% at layer 0) and saturate by layer 3; the pawn/rook square starts near chance (19%) and keeps improving through the last layer (64%); more depth is allocated to the piece that moves rarely. M3’s WDL/DTZ macro-F1 peaks at layers 3–4 and degrades by layer 5.

M4: the linear probe barely beats baseline out of range (+0.08 to +0.79); the MLP control does better (+0.39 to +1.70) but is 20–30 times worse than in-distribution, it is not extrapolating well.

The geometric baseline (Table 2) beats vanilla on static WDL (0.82 vs. 0.74) but is clearly beaten by every model probe on M4 generalization, where dynamics beyond static geometry evidently matter. The model’s advantage over raw position features is task-specific, not general.

In M5 the vanilla model shows a weak and diffuse effect with no layer standing out (median 0.07–0.21 across layers 2–5 over 250 pairs), reproduced across 2 seeds and 2 DTZ ranges. Multitask shows an almost perfect and concentrated effect at layer 5 (median 1.00, 88–90% of pairs > 0.5), exactly its supervised layer.

5. Conclusion

We can deduce that outcome-proximity representation is weak (M3), extrapolates poorly (M4) and lacks causal strength at tactical points (M5). Smooth tradeoffs recur: multitask training gains clarity at a cost to generalization (Table 2), while weight merging gains KRK competence at a cost to KPK (Figure 3), reflecting capacity allocation over duplication. Geometric baselines confirm this advantage is absent in unsupervised classification, but substantial

for distance generalization for look-ahead. In the future we can stratify M3/M4 by zugzwang-rate and test distributed information via multi-layer patching across layers 2–4.

6. Statement on the use of AI

I used Claude as a coding and writing assistant to debug code, accelerate the Google Colab environment setup and select the best libraries for the project. In this paper I used it to select the most relevant citations, after doing my independent review and reading of several papers on the topic.

References

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Appendix

A. Legibility vs. OOD Generalization Tradeoff (M3 and M4)

Figure 1 highlights the central tradeoff of multitask training: while auxiliary supervision drastically improves in-distribution WDL legibility (M3), it degrades out-of-distribution distance generalization (M4).

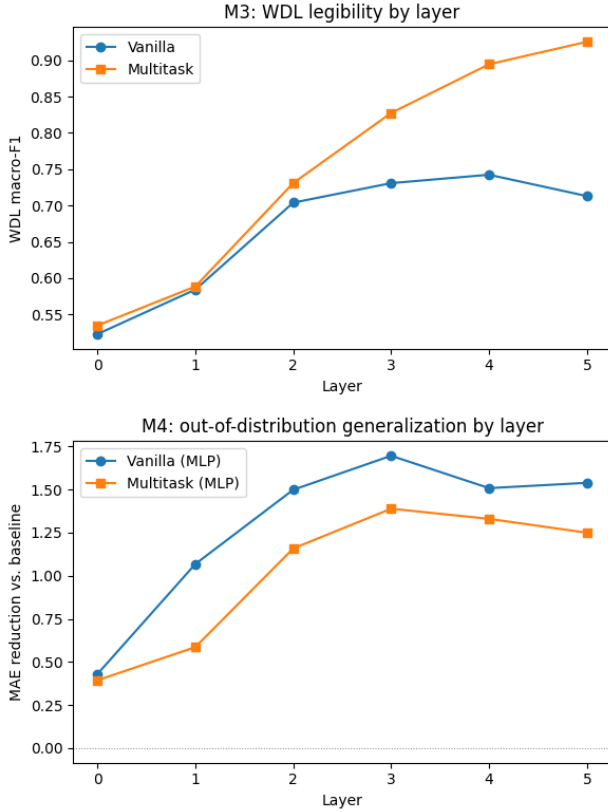


Figure 1. Left (M3): WDL macro-F1 across layers. Multitask achieves near-perfect legibility at layer 5. Right (M4): OOD distance generalization error reduction over baseline. Vanilla consistently outperforms Multitask in OOD extrapolation across deep layers.

B. Causal Activation Patching and Stratification (M5)

Table 3 and Figure 6 display the causal recovery metrics for M5.

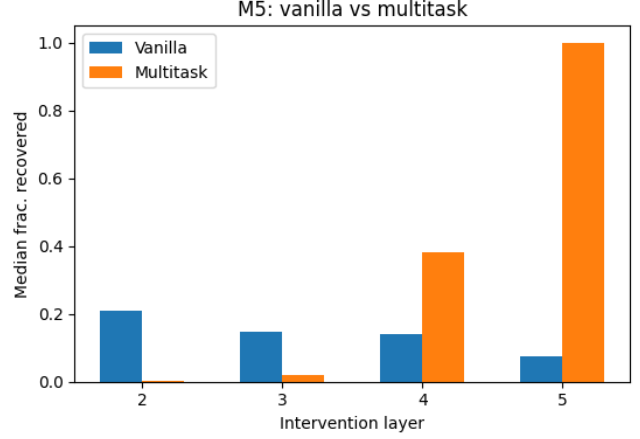


Figure 2. M5 Causal Patching. Left: Vanilla shows diffuse causal recovery across layers 2–5, whereas Multitask concentrates causality entirely at layer 5. Right: Stratification of Vanilla model shows weaker causal recovery on tactically sharp zugzwang positions compared to generic positions.

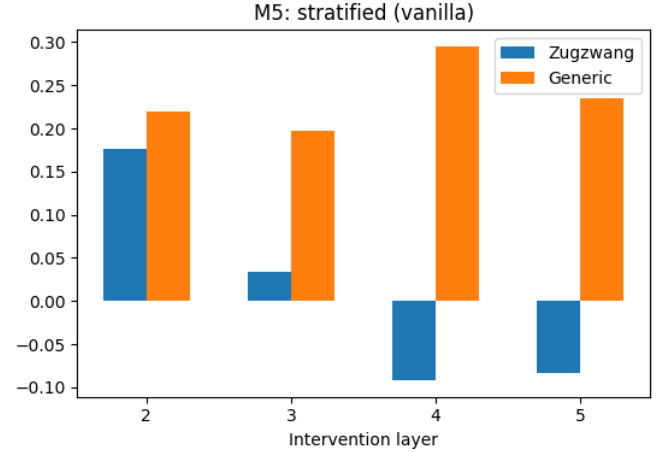


Table 3. M5 stratified by branching-position type (median recovery; 67 zugzwang vs. 183 generic pairs).

Layer	Vanilla		Multitask	
	zugzwang	generic	zugzwang	generic
2	0.18	0.22	0.03	0.00
3	0.03	0.20	0.01	0.02
4	-0.09	0.30	0.10	0.67
5	-0.08	0.24	0.91	1.00

C. Weight-Merging Task Vector Interpolation (M6)

Figure 3 and Table 4 demonstrate the task-vector merge $\theta(\alpha) = \theta_{base} + \alpha\tau$ between the base KPK model and the fine-tuned KRK specialist.

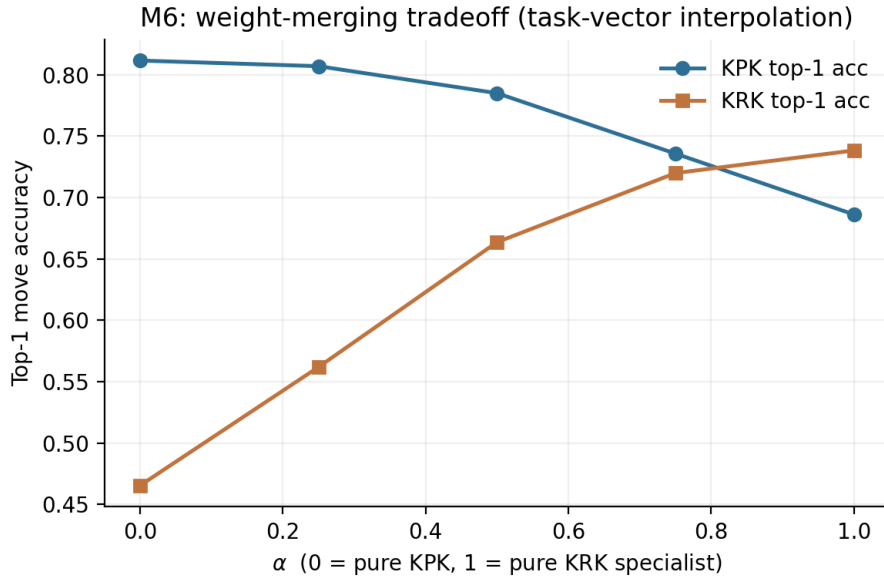


Figure 3. M6 Weight-merging accuracy tradeoff across interpolation coefficients $\alpha \in [0, 1]$. Performance degrades smoothly without catastrophic collapse.

Table 4. M6 task-vector merge accuracy values.

α	0.00	0.25	0.50	0.75	1.00
KPK acc	0.81	0.81	0.79	0.74	0.69
KRK acc	0.47	0.56	0.66	0.72	0.74